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cAMP-Stimulated Adenylate Cyclase Activation in *Dictyostelium discoideum* Is Inhibited by Agents Acting at the Cell Surface

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The ability of *Dictyostelium discoideum* amoebae to synthesize and secrete cAMP in response to exogenous cAMP is called cAMP signaling. Concanavalin A is a potent, rapid, noncompetitive inhibitor of this response, with the rate of inhibition consistent with its rate of binding. The concanavalin A does not deplete cellular ATP, alter cAMP binding to its surface receptors, or affect basal adenylate cyclase activity, but blocks the cAMP-stimulated activation of adenylate cyclase. Therefore, concanavalin A appears to inhibit a step between the receptor and the adenylate cyclase which is necessary for the transduction of the cAMP signal. Wheat germ agglutinin, a polyclonal antibody against an 80-kDa glycoprotein, four monoclonal antibodies against the amoebal surface, and a chemical cross-linking agent which reacts with cell surface primary amines also inhibit signaling. To determine the importance of cross-linking in the inhibition, succinylated concanavalin A and the unlinked, reactive portion of the chemical cross-linker were tested and found to be relatively ineffective inhibitors. Thus it appears that ligands capable of cross-linking molecules on the external surface of *D. discoideum* amoebae inhibit cAMP signaling. It is proposed that these cross-linking agents prevent membrane or cytoskeletal rearrangement and that this rearrangement must occur before the adenylate cyclase is activated. © 1984 Academic Press, Inc.

INTRODUCTION

Dictyostelium discoideum is an important model for studying cell–cell communication during development. In the life cycle of this organism, starved amoebae develop a system of chemical communication which consists of an adenylate cyclase, an external cAMP receptor, and phosphodiesterases (for a review see Devreotes, 1982). The binding of cAMP to its surface receptor transiently activates the adenylate cyclase (Roos and Gerish, 1976; Devreotes and Steck, 1979). The newly synthesized cAMP is secreted by the amoebae, while intracellular and extracellular phosphodiesterases act to return the cAMP concentration to its prestimulus level (Dinauer *et al.*, 1980a). This synthesis and secretion of cAMP in response to external cAMP is called signaling. With this communication system, a single amoeba which begins to secrete cAMP in a pulsatile manner can initiate cAMP waves which are relayed through the cell population (Tomchik and Devreotes, 1981). These waves plus an acquired chemotactic response to cAMP (Devreotes, 1982) result in the aggregation of 10^5 to 10^6 amoebae into a single multicellular unit.

Attempts to activate the adenylate cyclase of *D. discoideum in vitro* by the addition of cAMP have failed. Therefore a search for inhibitors of this process has been undertaken. A knowledge of how these inhibitors function may give an insight into the steps which lead to activation of the adenylate cyclase in *Dictyostelium.* Brenner and Thoms (1984) recently reported that caffeine specifically inhibits adenylate cyclase activation in these amoebae and have speculated that caffeine acts by altering the intracellular calcium distribution. Here we report that concanavalin A (Con A), a plant lectin which binds to glucose or mannose residues (Goldstein, 1976), also inhibits the cAMP-dependent stimulation of adenylate cyclase. Further studies suggest that Con A achieves this inhibition by cross-linking cell surface components.

MATERIALS AND METHODS

Cell growth, development, and perfusion assay. D. discoideum strain NC-4 was grown in association with *Enterobacter aerogenes* (Sussman, 1966) to a density of 5×10^8 cells/100 × 15-mm Petri plate. The amoebae were harvested and washed free of bacteria with development buffer (DB, 10 m*M* Na-K PO_4 buffer, pH 6.9, 1 m*M* $MgSO_4$, 0.2 m*M* $CaCl_2$). One milliliter of amoebae at 10^7/ml was shaken in DB in the presence of [^{3}H]adenosine-labeled bacteria (Devreotes *et al.*, 1979). After 3 hr the amoebae were allowed to settle on a 35 × 10-mm Petri plate containing 2% agar in DB and the fluid was removed. After developing for 3–4 hr more, the aggregating amoebae were resuspended in 2 ml DB and 200 μl placed on a Millipore filter (type ss 1300, Bedford, Mass.). These filters were part of a 4

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filter perfusion apparatus. When the effect of the monoclonal antibodies, 3,3′-dithiobis(sulfosuccinimidylpropionate) (DTSSP), and *N*-hydroxysulfosuccinimide (HOSu(SO_3)) was examined, the amoebae were shaken in the presence of the desired ligand for 30 min before being placed on the perfusion filter.

The perfusion assay was used to quantitate cAMP signaling (Devreotes *et al.*, 1979; Dinauer *et al.*, 1980b). Briefly, the [^{3}H]adenosine-labeled amoebae are perfused with buffer to remove the secreted phosphodiesterase and cAMP (Devreotes *et al.*, 1979). The buffer can rapidly be replaced with other solutions containing inhibitors, cAMP, etc. The secreted solutions are collected, phosphodiesterase is inactivated (Dinauer *et al.*, 1980b), and the secreted cAMP is removed from other labeled, secreted compounds by sequential dowex and alumina chromatography (Salomon, 1979). The cpm/fraction is proportional to the cAMP secretion rate; this rate is directly proportional to the intracellular cAMP concentration and reflects activation of the adenylate cyclase (Dinauer *et al.*, 1980a). With two 4 filter apparati, the relay response of control amoebae and those treated in 6 different manners can be determined.

cAMP binding assay. Amoebae were grown, developed (but not labeled with [^{3}H]adenosine), and placed on the perfusion filters. Half of the filters were perfused with 20 μg/ml Con A for 6–10 min while the controls received DB. Before adding the [^{3}H]cAMP, the filters were removed and placed on waxed paper to prevent fluid flow. [^{3}H]cAMP (10 n*M*) in DB containing 10 m*M* dithiothreitol was added to the filters and allowed to set for 30 sec. The [^{3}H]cAMP solution was drawn through by placing the filter on 2 layers of Whatman 3M paper. The amoebae were then washed twice with 0°C, saturated ammonium sulfate (van Haastert and Kein, 1983), the filter placed in 0.4 ml of 1% sodium lauryl sulfate, and scintillation fluid added. Nonspecific binding, that [^{3}H]cAMP remaining which was not bound to a receptor, was measured by adding 100 μM unlabeled cAMP to the tritiated solution. Specific binding was defined as cpm (total) − cpm (nonspecific). The total:nonspecific ratio was approximately 3:1.

Relative ATP levels. To determine relative levels of ATP, amoebae were developed in the presence of 100 μCi/ml of [^{32}P]PO_4 and placed on the perfusion filters as already described. Some amoebae were pretreated for 6 min with 20 μg/ml Con A, some with 1 m*M* sodium azide, and others with buffer alone. The filters were then inverted into Falcon multiwell tissue culture plates (Cockeysville, Md.) containing 300 μl ethanol (0°C). After 5 min, the ethanol/cell extract was centrifuged for 2 min in a microfuge and the pellet discarded. Aliquots of the supernatant were spotted on a PEI-cellulose thin layer chromatography plate and developed halfway in 0.5 *M* ammonium sulfate then placed in 0.7 *M* ammonium sulfate to complete development (Neuhard *et al.*, 1965). The plates were dried and autoradiographed. The ^{32}P spots which comigrated with an ATP standard were scraped into scintillation vials and counted.

Adenylate cyclase assay. Adenylate cyclase activity was measured by placing filters containing developed amoebae into 300 μl of a reaction mix containing 1 m*M* cAMP, 3 m*M* creatin phosphate, 0.25 m*M* ATP, 12.5 m*M* Tris pH 7.5, 1 mg/ml creatin phosphokinase, 12 m*M* CHAPS (3-[(3-cholamidopropyl)dimethylammonio]-1-propanesulfonate), 1 m*M* $MnCl_2$, 1 m*M* $MgCl_2$, 2 m*M* Na/K PO_4 and trace amounts of [^{32}P]ATP. The zwitterionic detergent, CHAPS, solubilizes the amoebae and allows the cyclase reaction to proceed (A. Theibert, personal communication). For activated adenylate cyclase, the amoebae were placed in the reaction mix 1 min after stimulation with 1 μM cAMP. The reactions were stopped after 1 min by the addition of 2% sodium lauryl sulfate, 9 m*M* ATP, and 1.3 m*M* cAMP (Salomon, 1979). The [^{32}P]cAMP was then separated from the other radiolabeled compounds (Salomon, 1979) and counted.

Materials. Wheat germ agglutinin was purchased from Miles (Elkhart, Ind.) and Vector (Burlingame, Calif.) and succinylated Con A from Vector. DTSSP and HOSu(SO_3) were obtained from Pierce (Rockford, Ill), the PEI-cellulose plates from Brinkmann (Westbury, N. Y.), and radioisotopes from ICN (Irvine, Calif.). All other chemicals were purchased from Sigma (St. Louis, Mo.).

RESULTS

A 6-min pretreatment with 10 μg/ml concanavalin A strongly inhibits the cAMP signaling response of aggregation-competent *D. discoideum* (Fig. 1). The inhibitory effect is prevented by preincubating the Con A with 50 m*M* α-methylmannoside, a sugar to which Con A binds with high affinity, suggesting that Con A binding to an amoebal surface component(s) is necessary for inhibition. Treatment of Con A-inhibited amoebae with 100 m*M* α-methylmannoside reverses the inhibition, presumably by displacing the Con A, indicating that the Con A treatment is not harmful to the amoebae. The dose–response curve reveals that Con A is a potent inhibitor with 50% inhibition occurring at 0.5 μg/ml (Fig. 2). When the inhibition by Con A is partial, the subsequent relay response has kinetics similar to the control (Fig. 1D, E is an example). To determine the time required for Con A to exert its effect, 100 μl of 100 μg/ml Con A was applied in a 3-

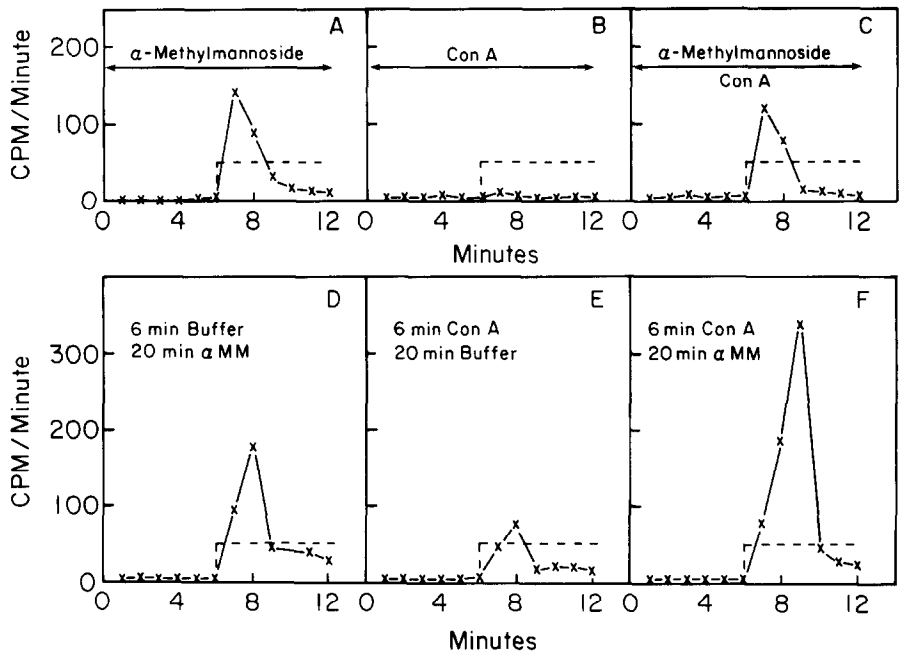

FIG. 1. The Con A inhibition is blocked by preabsorption with α-methylannoside (A, B, C) and can be reversed by application of this sugar (D, E, F). The rates of [^{3}H]cAMP secretion were measured by the perfusion technique. Plotted is the amount of [^{3}H]cAMP secreted in the previous minute (x). The dashed lines show where 1 μM cAMP was applied (A-F), with arrows indicating the presence of Con A (10 μg/ml) or α-methylmannoside (50 mM) (A, B, C). The amoebae (D, E, F) were pretreated as indicated with DB, 2 μg/ml Con A, or 100 mM α-methylmannoside, followed by 9 min with DB before the cAMP was applied.

sec pulse at varying times before cAMP addition (Fig. 3). If the Con A is applied with the cAMP stimulus 64% inhibition is observed, 80% inhibition occurring within the first 6 sec after Con A addition. This time course of inhibition is consistent with the time course for Con A binding to *Dictyostelium* (Darmon and Klein, 1976). If Con A is added while the amoebae are secreting cAMP, the response is abruptly inhibited (data not shown). The extent of inhibition caused by Con A is independent of the concentration of cAMP used to stimulate the amoebae (Table 1), indicating that the inhibition is noncompetitive.

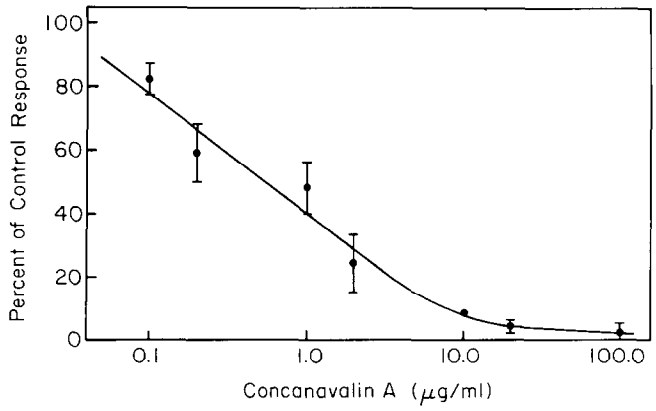

FIG. 2. Dose-response curve for Con A inhibition. The amoebae were pretreated for 6 min with the indicated concentration of Con A before being stimulated for 6 min with 1 μM cAMP. The control did not receive Con A. The response is measured by integrating under curves such as those in Fig. 1. The error bars represent standard deviation with a minimum of three replicates per point.

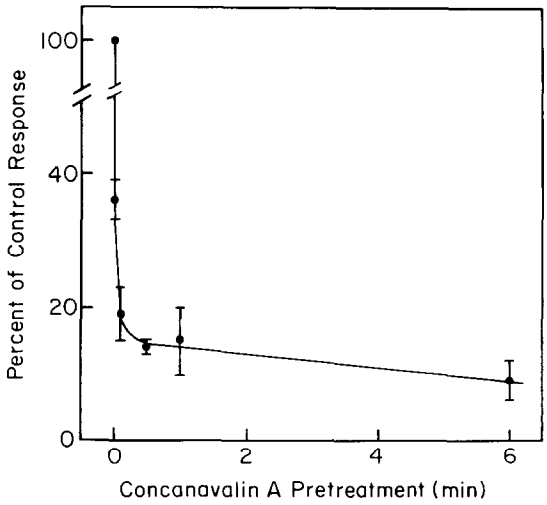

FIG. 3. Con A is a rapid inhibitor of signal relay: 100 μl of 100 μg/ml Con A was added in a 3-sec pulse at the times indicated. Zero time represents the addition of Con A with the 1 μM cAMP stimulus. The control response of 100% where 100 μl of DB is added, is also shown. The error bars represent standard deviation with a minimum of three replicates per point.

TABLE 1
INHIBITION BY CONCANAVALIN A IS NONCOMPETITIVE[a]

cAMP concentration (*M*)	Percentage control response[b]
10^{-8}	45 ± 3[c]
10^{-7}	44 ± 2
10^{-6}	48 ± 8
10^{-5}	47 ± 6

[a] The amoebae were incubated with 1 μg/ml Con A for 6 min prior to the cAMP stimulation.
[b] The response is quantitated by integrating under curves such as those in Fig. 1.
[c] The error represents standard deviation.

The above results showed that Con A is a potent, rapid, noncompetitive inhibitor of cAMP signaling in *D. discoideum*. Attempts were then made to determine the point of inhibition. Specific cAMP binding (10 n*M* cAMP) was measured and the untreated control cells bound 9.8×10^3 molecules/cell while the Con A-treated cells (20 μg/ml, 6 min) bound 9.5×10^3 molecules/cell. Juliani and Klein (1977) have reported a slight increase in cAMP binding after a brief Con A treatment, but this was not observed. van Haastert and Kien (1983), by examining the effects of a series of cAMP analogs on cAMP binding, chemotaxis, and guanylate cyclase activation, observed that the receptors detected in the binding assay have the same specificity and affinity as those mediating chemotaxis and guanylate cyclase activation. Theibert and Palmisano have evidence, using these analogs, that the cAMP surface receptors responsible for adenylate cyclase activation have the same specificity and affinity as those described by van Haastert and Kien (personal communication). These results imply one class of cAMP surface receptors. Because Con A does not inhibit cAMP-induced amoebal shape changes (data not shown), does not block the cAMP-induced guanylate cyclase activation (Mato *et al.*, 1978), and there is no demonstrable effect of Con A on cAMP binding, the Con A inhibition of signaling is probably not the result of Con A preventing cAMP from binding to its surface receptor.

In *D. discoideum* the addition of cAMP results in a rise in the intracellular level of cAMP prior to cAMP secretion (Dinauer *et al.*, 1980a), and Con A prevents this rise (Fig. 4). The absence of a rise in intracellular cAMP could be the result of (1) ATP depletion, ATP being the substrate for adenylate cyclase, (2) an activation of an intracellular phosphodiesterase such that the cAMP is degraded shortly after its synthesis, or (3) a blockage at an intermediate step linking the cAMP surface receptor and the adenylate cyclase thus preventing activation of the enzyme. Con A appears not to affect ATP levels or basal adenylate cyclase activity (Table 2). However when Con A is bound to the surface of an amoebae, binding of cAMP to its receptor does not result in activation of the adenylate cyclase (Table 2).

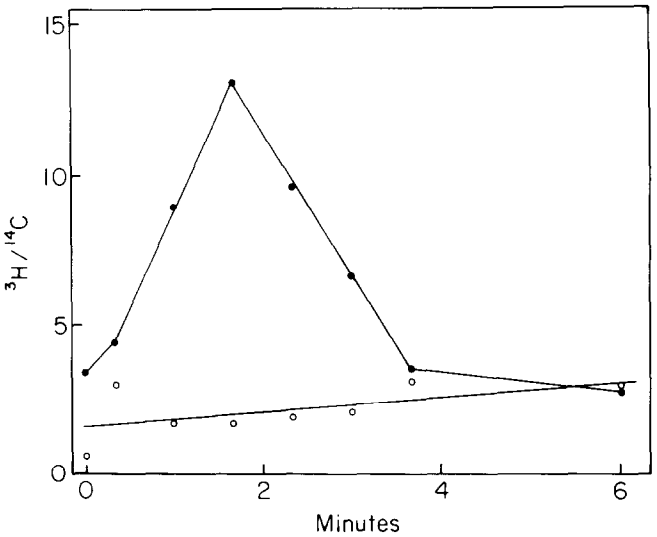

FIG. 4. Con A blocks the cAMP-induced rise in intracellular cAMP. cAMP (1 μ*M*) was applied, at time zero, to [^{3}H]adenosine-labeled amoebae which were being perfused. At the times indicated, a filter holding amoebae was inverted into a stop solution containing 8% formic acid, 1 m*M* cAMP, and 100 [^{14}C]cAMP. The cAMP was then separated from other radioactive compounds (Salomon, 1979) and the $^3H/^{14}C$ ratio determined. Half the amoebae were pretreated for 6 min with 20 μg/ml Con A before the cAMP stimulus (○), while the other half received DB (●). A typical experiment is shown, with similar results being obtained in four replicates.

Since no component of the transduction pathway between the cAMP surface receptor and the adenylate cyclase has been identified, the specificity of the lectin-induced inhibition of cAMP signaling was examined with the hope of identifying such a molecule. The tetravalent lectin, wheat germ agglutinin (WGA), has been shown to bind to *D. discoideum* (West *et al.*, 1978; Burridge and Jordan, 1979) and WGA proved to be an

TABLE 2
POSITION OF CONCANAVALIN A INHIBITION

Treatment	Relative ATP levels	Basal cyclase (pmole/min × 10^7 cells)	Activated cyclase (pmole/min × 10^7 cells)
Buffer	1.0	2.6 ± 0.6[a]	6.5 ± 0.8
Con A[b]	1.2 ± 0.3	1.9 ± 0.9	2.0 ± 0.5
Azide[c]	0.7 ± 0.2	n.d.[d]	n.d.

[a] The error represents standard deviation.
[b] The amoebae were pretreated for 6 min with 20 μg/ml Con A.
[c] The amoebae were pretreated for 6 min with 1 m*M* NaN_3.
[d] Not determined.

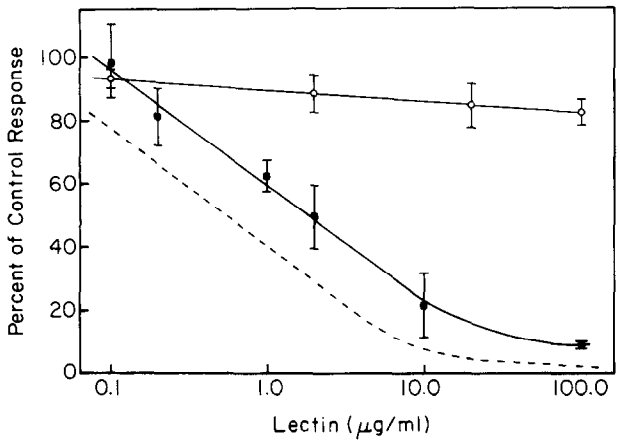

FIG. 5. The dose-response curves for WGA (■) and succinylated Con A (○) are compared with that of Con A (- -), Con A data replotted from Fig. 2. The amoebae were pretreated for 6 min with the indicated concentration of lectin before being stimulated for 6 min with 1 μM cAMP. The controls did not receive the lectin. The response is quantitated by integrating under curves such as those in Fig. 1. The bars represent standard deviation with a minimum of three replicates per point.

effective inhibitor of signaling with 50% inhibition occurring at 2 μg/ml (Fig. 5). When WGA was applied to 5×10^6 amoebae/ml in suspension, 65 μg/ml yielded 50% inhibition.[3] Since Con A and WGA are potent inhibitors of cAMP relay, either they both interact with a critical component(s) involved in signaling or possibly the mere presence of a ligand inhibits the *D. discoideum* cAMP signaling response.

Antibodies were used to further examine whether binding to a specific surface component is necessary to exert inhibition. A polyclonal antibody against a putative *D. discoideum* adhesion molecule, a glycoprotein with a molecular weight of 80,000 (gp 80), was obtained (gift of Dr. Bill Loomis). Western blot analysis against an extract of developed amoebae revealed that this antibody binds to at least 8 proteins (Murray *et al.*, 1981). When 50 μg/ml of the antibody was perfused past developed amoebae for 6 min, subsequent cAMP stimulation resulted in only 54% of the control response (Table 3). Four monoclonal antibodies against the surface of *D. discoideum* were also obtained (gift of Alan Smith and Dr. Steve Barclay). When amoebae are pretreated with these antibodies, the relay response is inhibited (Table 3). These monoclonal antibodies are not fully characterized, but it is known that there is no membrane protein which is bound by all four antibodies (S. Barclay, personal communication).

[3] When the lectins are perfused past the amoebae their binding is pseudo-first order because the lectin concentration in the perfusing medium is constant. In solution the effective concentration of the lectin decreases as it binds to the amoebae, thereby shifting the dose-response curve and making it dependent upon cell number.

To examine the role of valency on the inhibition of cAMP relay, succinylated Con A (succinyl-Con A) was tested. Succinylation of Con A results in a dimeric molecule with unaltered affinity for α-methylglucose and an unaltered ability to bind to cell surface receptors (Gunther *et al.*, 1973). Because of the reduced valency, succinyl-Con A has a drastically reduced ability to carry out functions which involve aggregation of its receptors (for a review see Wang *et al.*, 1976). Consistent with these observations, Weeks and Weeks (1975) have shown that succinyl-Con A, which does not cause cell agglutination, can block Con A-induced cell-cell agglutination in *D. discoideum*. Succinyl-Con A is relatively ineffective as an inhibitor of cAMP signaling (Fig. 5), with a pretreatment of 100 μg/ml for 6 min inhibiting the response by only 12%.

Since the results with Con A, succinyl-Con A, WGA, and the antibodies suggest that the ability to inhibit the relay response appears to be related to valency and not to the identity of the bound molecules, the effect of the chemical cross-linking agent, DTSSP, was examined. This agent is divalent and interacts with primary amines on the cell surface (Starvos, 1982). DTSSP inhibited the cAMP relay response by 86% at 10 mM (Fig. 6). If the amoebae are treated with the unlinked reactive portions of this molecule, ($HOSu(SO_3)$), only 9% inhibition is observed at twice this concentration. These results suggest that the mode of inhibition is cross-linking surface components.

TABLE 3
INHIBITION BY ANTIBODIES

Antibody	Type	Concentration	Percentage control response[a]
Anti-gp80[b]	Polyclonal	50 μg/ml	54
2D7	Monoclonal[c]	20 μl/ml	60
		40	41
2B8	Monoclonal	20	71
		40	69
4G4	Monoclonal	20	76
		40	58
H11	Monoclonal	20	100
		40	85

[a] The anti-gp80 was in 14 mM NaCl, so the control amoebae were perfused with 14 mM NaCl in DB. For the monoclonal antibody experiments, the control amoebae were pretreated with hybridoma growth medium. The response in quantitated by integrating under curves such as those in Fig. 1.

[b] The anti-gp80, IgG, was perfused past the amoebae.

[c] The supernatant from hybridoma cell cultures was added in all monoclonal antibody experiments.

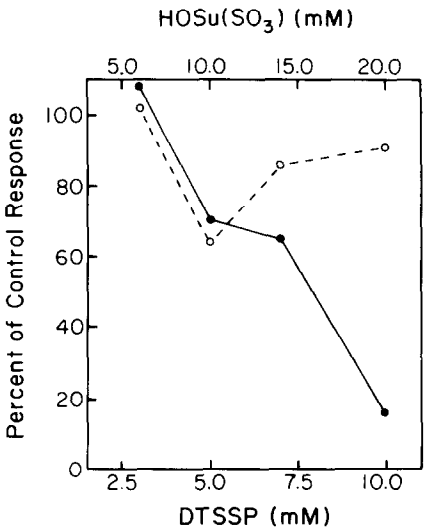

FIG. 6. The dose-response curves for the cross-linking agent, DTSSP, and its reactive group HOSu(SO_3). The amoebae were pretreated for 30 min, in solution, with the indicated concentration before being placed in the perfusion device. The amoebae were then perfused with buffer for 30 min before 1 μM cAMP was applied for 6 min. The concentration of HOSu(SO_3) being compared with DTSSP is twice that of DTSSP because each molecule of DTSSP contains two molecules of HOSu(SO_3). The points represent at least duplicate determinations.

DISCUSSION

Several investigators have shown that *D. discoideum* development is affected by the addition of the lectin concanavalin A. Low concentrations of Con A (less than 50 μg/ml with cells at 2×10^7/ml) stimulate development (Darmon and Klein, 1976) whereas higher concentrations delay or block it (Gillette and Filosa, 1973; Weeks and Weeks, 1975; Darmon and Klein, 1976). High concentrations of Con A have been shown to inhibit the cAMP chemotactic response (Gillette and Filosa, 1973), to cause a premature rise in cellular phosphodiesterase (Gillette and Filosa, 1973; Darmon and Klein, 1976), and to increase the level of an extracellular phosphodiesterase inhibitor (Darmon and Klein, 1976). Because the amoebae bind and internalize Con A (Condeelis, 1979) or extrude their bound Con A receptors into the medium (Gillette *et al.*, 1974; Hellio and Ryter, 1980) and because these responses or the assays for them are slow, the concentrations of Con A necessary to exert the previously reported effects is uncertain. The only rapid responses to cAMP which have been measured in the presence of Con A, the cAMP-induced rise in cGMP and the cAMP-induced decrease in light scattering, were unaffected by Con A at 10 μg/ml (treated in solution) and at 100 μg/ml the rise in cGMP was enhanced whereas the light scattering response was inhibited (Mato *et al.*, 1978).

Con A, WGA, DTSSP, anti-gp80, and 4 different monoclonal antibodies inhibit the *D. discoideum* cAMP signaling response. Although Con A and WGA recognize surface carbohydrates, they differ in specificity with Con A preferring mannose and glucose residues and WGA preferring (*N*-acetylglucosamine)$_2$ (Lis and Sharon, 1977). On *D. discoideum* most of the receptors for WGA are distinct from those for Con A (Molday *et al.*, 1976; West *et al.*, 1978; Hellio and Ryter, 1980). Since all 5 antibodies tested were also inhibitory, it is unlikely that the binding to a single surface component blocks the signaling response.

The results obtained with Con A versus succinyl-Con A, and DTSSP versus HOSu(SO_3) indicate that cross-linking surface components causes inhibition. If the level of inhibition is controlled by the degree of cross-linking of surface components, it is possible that the monoclonal antibodies did not have enough binding sites to affect full inhibition. This would explain the apparent saturation of the monoclonal antibody-induced inhibition at less than 100%. Con A and WGA would have sufficient binding sites to induce near-complete inhibition (Weeks, 1975; Darmon and Klein, 1976; West and McMahon, 1977). DTSSP, which reacts with primary amines on the cell surface (Starvos, 1982), should also be capable of extensive cross-linking. Although Con A and antibody-induced cross-linking often leads to capping, the rate at which the Con A inhibits signaling is too fast for Con A-induced capping to be involved (Gillette *et al.*, 1974; Condeelis, 1979; Hellio and Ryter, 1980). This conclusion is supported by the WGA-induced inhibition; WGA does not cause capping in *D. discoideum* (Molday *et al.*, 1976; Hellio and Ryter, 1980).

Altering the surface, apparently by cross-linking, blocks the flow of information from the occupied cAMP surface receptor to the adenylate cyclase, thus preventing activation of this enzyme without affecting its basal activity. It has been shown that during cAMP signaling the number of vesicles and empty vacuoles within the amoebae increases and the cell surface becomes highly irregular whereas before the response it was smooth (Maeda and Gerisch, 1977). The time course of the rise in vesicles is consistent with these structures being involved in adenylate cyclase activation. Studies with Con A, WGA, and cationized ferritin have shown that as long as these ligands occupy the majority of the *D. discoideum* amoebal surface phagocytosis cannot occur, with particle adsorption and membrane fusion blocked (Hellio and Ryter, 1980). If membrane shuttling, e.g., endocytosis or exocytosis, is a vital part of signaling, as suggested by the electron microscopy results, it is reasonable that its blockage would eliminate the response. Alternatively the cross-linking agents may be affecting the cytoskeleton or

blocking a necessary lateral or transverse rearrangement of molecules in the plasma membrane. The results discussed in this paper are consistent with these interpretations, but further experiments are needed to determine the mechanism of action of these inhibitors.

In *D. discoideum* both cell-cell contact and cAMP regulate gene expression (for a review see Lodish *et al.*, 1982). The results reported in this paper hint at a possible link between cell-cell contact and cAMP levels. That is, if cell-cell contact causes cross-linking of cell surfaces, it is possible that this contact modulates adenylate cyclase activity thereby altering the cAMP level in the cell.

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